

Module 4 — Quiz B (Concept Check)

Question bank · After Read 2 · open notes · unlimited attempts

Module 4, Quiz B: Concept check

Reading: [Module 4 Read 2](#) (complete Read 1 + Quiz A first)

Canvas: Concept Check quiz (Module 4)

Apply design classification, methods-map matching, and R output reading from Read 1–2.

Full integration, simulation, and error scenarios → [Synthesis](#).

Section A — Design classification (Read 1; revisit with R in mind)

A1. Exercise trial

60 volunteers randomly assigned to aerobic exercise or stretching for 8 weeks; resting heart rate measured.

1. RCT or observational? 2. Can we make a causal claim? 3. What is the confounder risk?

A2. Diet survey

Survey of 200 adults; self-reported diet quality and cholesterol level compared.

1. RCT or observational? 2. Causal claim? 3. Confounder risk?

A3. Classroom randomization

Schools randomly assigned to new or traditional math curriculum; end-of-year scores compared.

1. RCT or observational? 2. Unit of randomization? 3. Confounder risk?

A4. Coffee and productivity

Employees who drink 2 cups/day report higher productivity. Identify a likely confounder and explain the DAG path.

Answer key — Section A — Design classification (Read 1; revisit with R in mind)

1. 1. **RCT**. 2. **Yes** — random assignment supports causation. 3. **Low** — randomization distributes confounders across groups.
 2. 1. **Observational**. 2. **No** — association only; diet is self-selected. 3. **High** — age, income, lifestyle are likely confounders.
 3. 1. **RCT** (cluster-randomized by school). 2. **School** (not individual student). 3. **Low** at school level — within-school confounders remain.
 4. Example: **sleep quality** — people who sleep well drink more coffee AND report higher productivity. DAG: sleep → coffee (EV) and sleep → productivity (RV).
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Section B — Methods-map matching (Read 2 §1)

B1. Two numerical groups

Two independent groups; outcome is **blood pressure** (numerical). Which test and R function?

B2. Two binary groups

Two groups; outcome is **recovered** (yes/no). Sample sizes $n_1 = 80$, $n_2 = 75$. Which test and R function?

B3. Matched pairs vs independent

Same students tested before and after an intervention. Which design — matched pairs or independent groups? Which test?

B4. Expected counts condition

When using `chisq.test()`, what condition must you check?

B5. Welch vs equal-variance t

Why does R's `t.test()` use the **Welch** approximation by default (not equal-variance t)?

Answer key — Section B — Methods-map matching (Read 2 §1)

1. **Welch two-sample t -test**. R: `t.test(bp ~ group, data = df)`. Conditions: independence, reasonably normal or large n per group.
 2. **Two-proportion z-test** or **chi-square test of independence**. R: `prop.test(c(x1, x2), c(80, 75))` or `chisq.test(table(df$group, df$recovered))`.
 3. **Matched pairs** (same units, two measurements). Use a **paired t -test** on differences: `t.test(df$post - df$pre)` or `t.test(post, pre, paired = TRUE)`.
 4. All **expected cell counts** 5 (or 1 by relaxed rule). If violated, use Fisher's exact test or combine categories.
 5. **Welch** does not assume equal population variances across groups — more robust and appropriate unless you have strong evidence the variances are equal.
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Section C — Read two-group R output (Read 2 §1)

C1. Welch t -test decision

Welch two-sample t -test: $p = 0.03$, 95% CI [1.2, 8.6] (Group B – Group A). State the decision ($\alpha = 0.05$) and interpret the CI.

C2. Fail to reject

Two-proportion z -test: $p = 0.18$, 95% CI [−0.02, 0.10] (proportion difference). State the decision ($\alpha = 0.05$).

C3. Identify null and alternative

Two groups; H_0 : mean blood pressure is the same in both groups. Write the null and alternative hypotheses using μ_1 and μ_2 .

Answer key — Section C — Read two-group R output (Read 2 §1)

1. **Reject** H_0 ($p < 0.05$). We are 95% confident the population mean for Group B exceeds Group A by between 1.2 and 8.6 units.
 2. **Fail to reject** H_0 ($p > 0.05$). Insufficient evidence of a difference in population proportions.
 3. $H_0 : \mu_1 = \mu_2$ (or $\mu_1 - \mu_2 = 0$) vs $H_a : \mu_1 \neq \mu_2$ (two-sided) or directed alternative if specified.
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